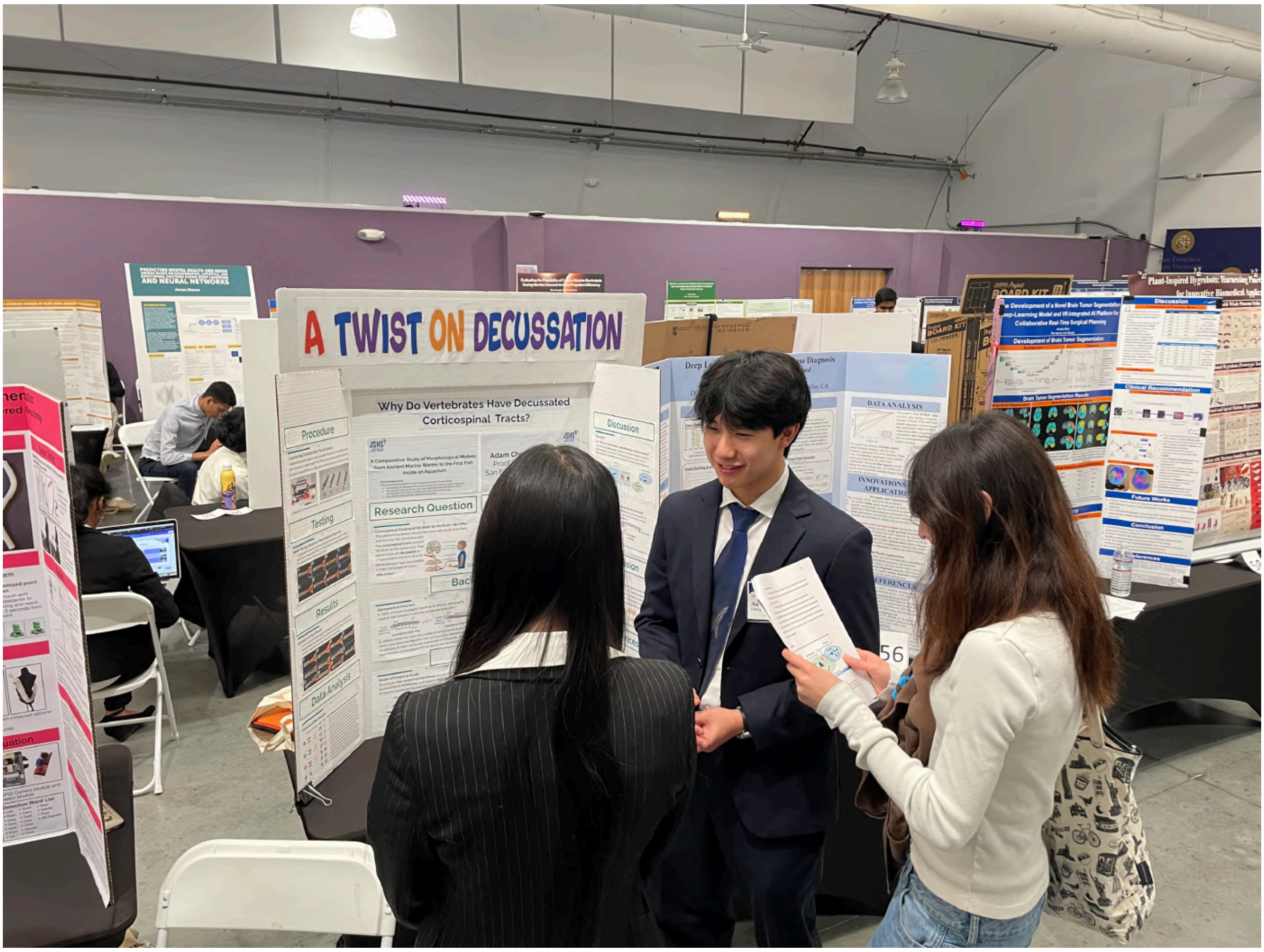
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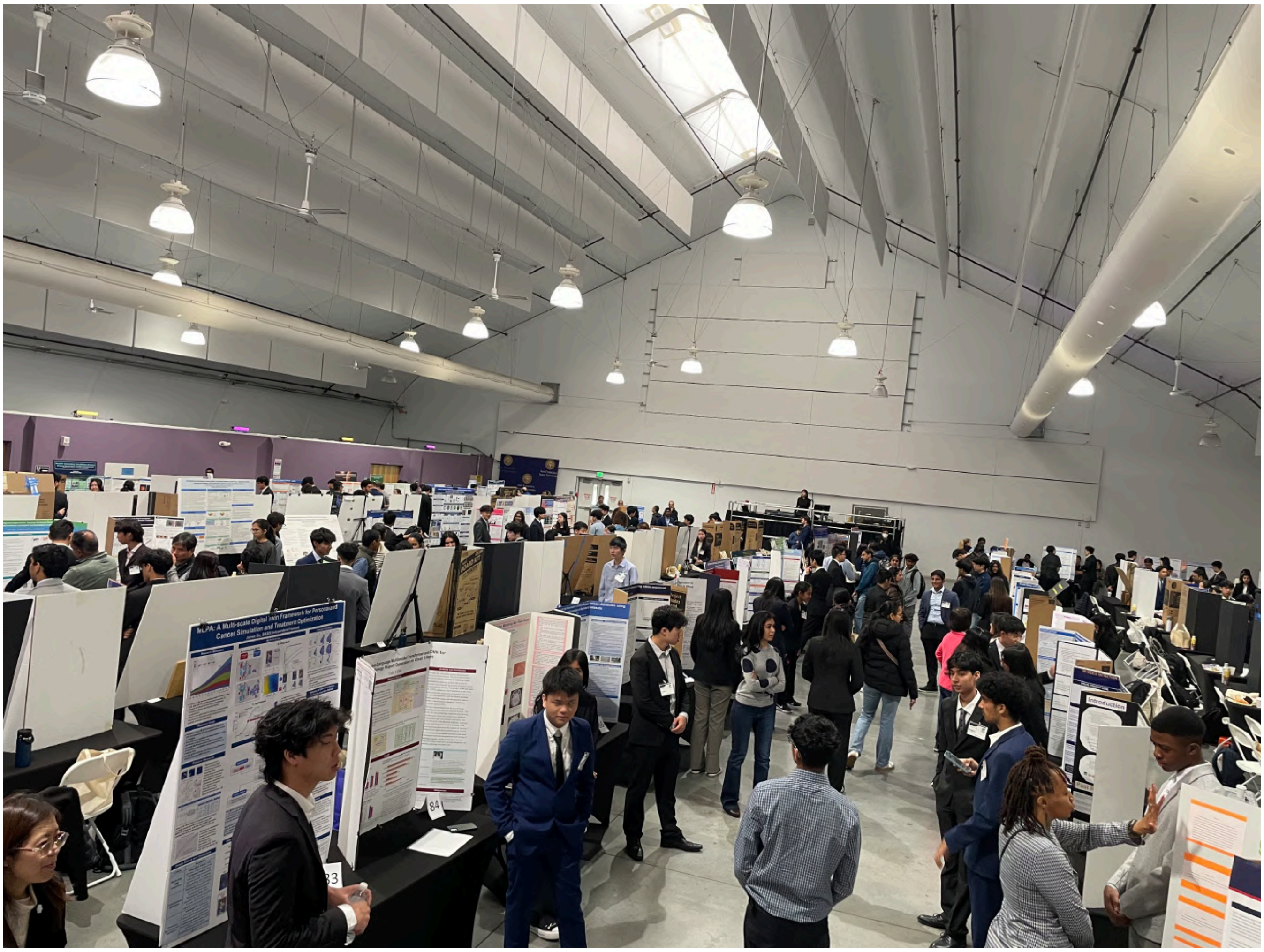
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# “Why Do Vertebrates Have Decussated Corticospinal Tracts?” — A Comparative Study of Morphological Models From Ancient Marine Worms to the First Fish Inside an Aquarium

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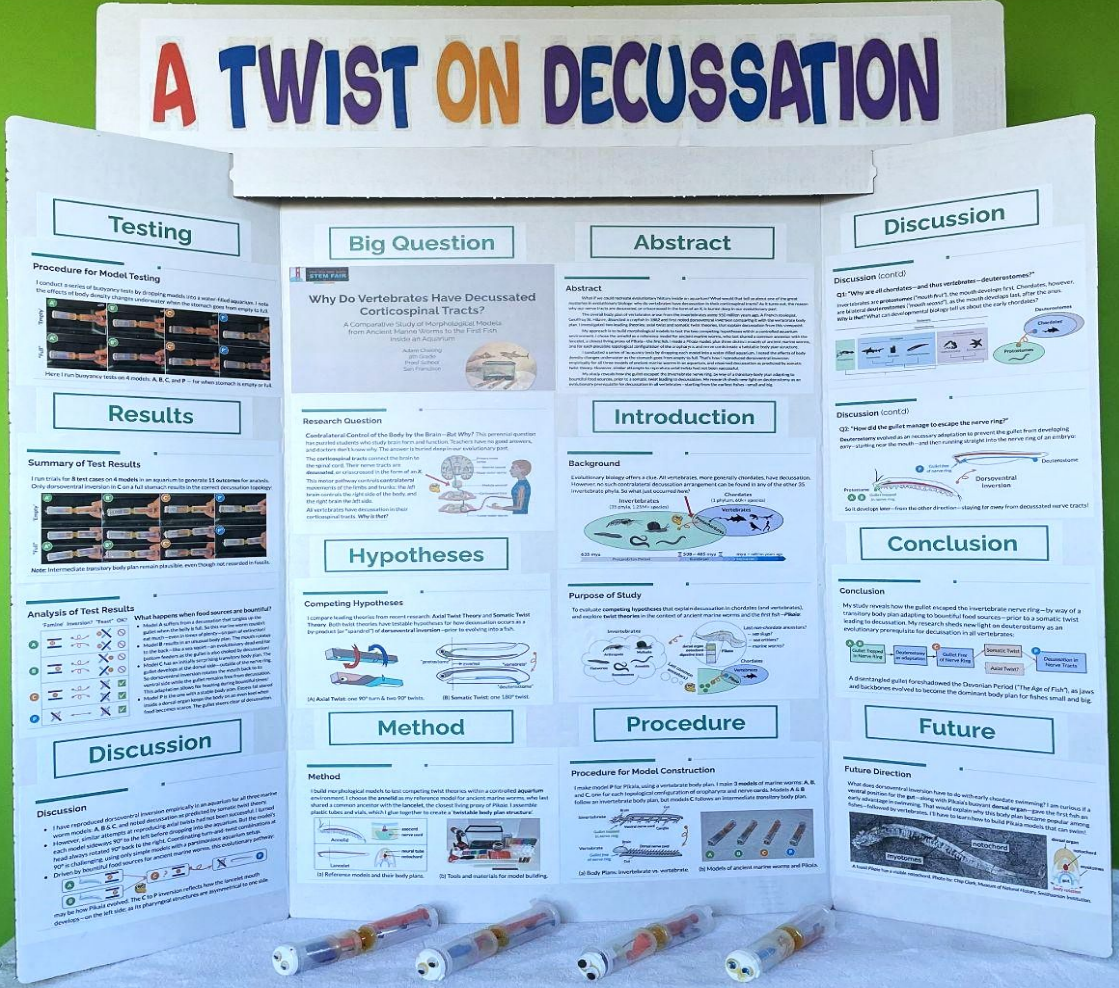
What if we could recreate evolutionary history inside an aquarium? What would that tell us about the greatest mystery in evolutionary biology: why do vertebrates have decussation in their corticospinal tracts?

The reason why our nerve tracts are *decussated* is buried deep in our evolutionary past. I investigated two leading theories: axial twist theory and somatic twist theory, that describe how the vertebrate body plan arose from the invertebrates 550 million years ago.

To test the two competing hypotheses, I built morphological models for empirical study within an aquarium environment. I chose the annelid as a reference model for ancient marine worms, who last shared a common ancestor with the lancelet, a closest living proxy of *Pikaia*. I constructed three ancient marine worm models with distinct topological configuration of the oropharynx and nerve cords inside a twistable body plan structure, alongside a Pikaia model.

I conducted buoyancy tests by dropping each model into the aquarium. The effects of body density changes underwater were noted. When the stomach goes from empty to full, I observed decussation resulting from dorsoventral inversion as predicted by somatic twist theory. However, all attempts to reproduce axial twists had failed.

My study reveals how the gullet escaped the invertebrate nerve ring, through a *transitory* body plan adapting to bountiful food sources, followed by a *somatic twist* leading to decussation. My investigation uncovers an evolutionary role for *deuterostomy* that leads to decussated nerve tracts in vertebrates, and opens up new pathways for future research in evolutionary biology.



# A TWIST ON DECUSSATION

## Testing

### Procedure for Model Testing

I conduct a series of buoyancy tests by dropping models into a water-filled aquarium. I note the effects of body density changes underwater when the stomach goes from empty to full.

Here I run buoyancy tests on 4 models: A, B, C, and P — for when stomach is empty or full.

## Results

### Summary of Test Results

I run trials for **8 test cases** on **4 models** in an aquarium to generate **11 outcomes** for analysis. Only dorsoventral inversion in C on a full stomach results in the correct decussation topology.

**Note:** Intermediate transitory body plan remain plausible, even though not recorded in fossils.

### Analysis of Test Results

**What happens when food sources are bountiful?**

- Model A suffers from a decussation that tangles up the gullet when the belly is full. So this marine worm couldn't eat much—even in times of plenty—on pain of extinction!
- Model B results in an unusual body plan. The mouth rotates to the back—like a sea squirt—an evolutionary dead end for bottom feeders as the gullet is also choked by decussation!
- Model C has an initially surprising transitory body plan. The gullet develops at the dorsal side—outside of the nerve ring. So dorsoventral inversion rotates the mouth back to its ventral side while the gullet remains free from decussation. This adaptation allows for feasting during bountiful times!
- Model P is the one with a stable body plan. Excess fat stored inside a dorsal organ keeps the body on an even keel when food becomes scarce. The gullet steers clear of decussation.

## Discussion

### Discussion

- I have reproduced dorsoventral inversion empirically in an aquarium for all three marine worm models: A, B & C, and noted decussation as predicted by somatic twist theory.
- However, similar attempts at reproducing axial twists had not been successful. I turned each model sideways 90° to the left before dropping into the aquarium. But the model's head always rotated 90° back to the right. Coordinating turn-and-twist combinations at 90° is challenging, using only simple models with a parsimonious aquarium setup.
- Driven by bountiful food sources for ancient marine worms, this evolutionary pathway may be how Pikaia evolved. The C to P inversion reflects how the lancelet mouth develops—on the left side; as its pharyngeal structures are asymmetrical to one side.

## Big Question

STEM FAIR

### Why Do Vertebrates Have Decussated Corticospinal Tracts?

A Comparative Study of Morphological Models from Ancient Marine Worms to the First Fish Inside an Aquarium

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### Research Question

**Contralateral Control of the Body by the Brain—*But Why?*** This perennial question has puzzled students who study brain form and function. Teachers have no good answers, and doctors don't know why. The answer is buried deep in our evolutionary past.

The **corticospinal tracts** connect the brain to the spinal cord. Their nerve tracts are **decussated**, or crisscrossed in the form of an X.

This motor pathway controls **contralateral** movements of the limbs and trunks: the *left* brain controls the *right* side of the body, and the right brain the *left* side.

All vertebrates have decussation in their corticospinal tracts. **Why is that?**

## Hypotheses

### Competing Hypotheses

I compare leading theories from recent research: **Axial Twist Theory** and **Somatic Twist Theory**. Both twist theories have testable hypotheses for how decussation occurs as a by-product (or "spandrel") of **dorsoventral inversion**—prior to evolving into a fish.

(A) Axial Twist: one 90° turn & two 90° twists. (B) Somatic Twist: one 180° twist.

## Method

### Method

I build morphological models to test competing twist theories within a controlled **aquarium** environment. I choose the **annelid** as my reference model for ancient marine worms, who last shared a common ancestor with the **lancelet**, the closest living proxy of Pikaia. I assemble plastic tubes and vials, which I glue together to create a '**twistable body plan structure**'.

(a) Reference models and their body plans. (b) Tools and materials for model building.

## Abstract

### Abstract

What if we could recreate evolutionary history inside an aquarium? What would that tell us about one of the great mysteries in evolutionary biology: why do vertebrates have decussation in their corticospinal tracts? As it turns out, the reason why our nerve tracts are decussated, or crisscrossed in the form of an X, is buried deep in our evolutionary past.

The overall body plan of vertebrates arose from the invertebrates some 550 million years ago. A French zoologist, Geoffroy St. Hilaire, dissected a crayfish in 1822 and first noted dorsoventral inversion comparing it with the vertebrate body plan. I investigated two leading theories: axial twist and somatic twist theories, that explain decussation from this viewpoint.

My approach is to build morphological models to test the two competing hypotheses within a controlled aquarium environment. I chose the annelid as a reference model for ancient marine worms, who last shared a common ancestor with the lancelet, a closest living proxy of Pikaia—the first fish. I made a Pikaia model, plus three distinct models of ancient marine worms, one for each plausible topological configuration of the oropharynx and nerve cords inside a twistable body plan structure.

I conducted a series of buoyancy tests by dropping each model into a water-filled aquarium. I noted the effects of body density changes underwater as the stomach goes from empty to full. That's how I reproduced dorsoventral inversion empirically for all three models of ancient marine worms in an aquarium, and observed decussation as predicted by somatic twist theory. However, similar attempts to reproduce axial twists had not been successful.

My study reveals how the gullet escaped the invertebrate nerve ring, by way of a transitory body plan adapting to bountiful food sources, prior to a somatic twist leading to decussation. My research sheds new light on deuterostomy as an evolutionary prerequisite for decussation in all vertebrates—starting from the earliest fishes—small and big.

## Introduction

### Background

Evolutionary biology offers a clue. All vertebrates, more generally chordates, have decussation. However, no such contralateral decussation arrangement can be found in any of the other 35 invertebrate phyla. So what just occurred *here*?

### Purpose of Study

To evaluate **competing hypotheses** that explain decussation in chordates (and vertebrates), and explore **twist theories** in the context of ancient marine worms and the first fish—***Pikaia***.

## Procedure

### Procedure for Model Construction

I make model P for Pikaia, using a vertebrate body plan. I make **3 models** of marine worms: A, B, and C, one for each topological configuration of oropharynx and nerve cords. Models A & B follow an invertebrate body plan, but models C follows an intermediate transitory body plan.

(a) Body Plans: invertebrate vs. vertebrate. (b) Models of ancient marine worms and Pikaia.

## Discussion

### Discussion (contd)

**Q1: "Why are all chordates—and thus vertebrates—deuterostomes?"**

Invertebrates are **protostomes** ("mouth first"), the mouth develops first. Chordates, however, are bilateral **deuterostomes** ("mouth second"), as the mouth develops last, after the anus. **Why is that?** What can developmental biology tell us about the early chordates?

### Discussion (contd)

**Q2: "How did the gullet manage to escape the nerve ring?"**

**Deuterostomy** evolved as an necessary adaptation to prevent the gullet from developing early—starting near the mouth—and then running straight into the nerve ring of an embryo:

So it develops later—from the other direction—staying far away from decussated nerve tracts!

## Conclusion

### Conclusion

My study reveals how the gullet escaped the invertebrate nerve ring—by way of a transitory body plan adapting to bountiful food sources—prior to a somatic twist leading to decussation. My research sheds new light on deuterostomy as an evolutionary prerequisite for decussation in all vertebrates:

A
B
Gullet Trapped in Nerve Ring
Deuterostomy an adaptation
C
Gullet Free of Nerve Ring
Somatic Twist
Axial Twist?
P
Decussation in Nerve Tracts

A disentangled gullet foreshadowed the Devonian Period ("The Age of Fish"), as jaws and backbones evolved to become the dominant body plan for fishes small and big.

## Future

### Future Direction

What does dorsoventral inversion have to do with early chordate swimming? I am curious if a **ventral** position for the **gut**—along with Pikaia's buoyant **dorsal organ**—gave the first fish an early advantage in swimming. That would explain why this body plan became popular among fishes—followed by vertebrates. I'll have to learn how to build Pikaia models that can swim!

A fossil Pikaia has a visible notochord. Photo by: Chip Clark, Museum of Natural History, Smithsonian Institution.